



BACHELORS IN ARTIFICIAL INTELLIGENCE-BS(AI)

Structure of BS(AI) Programs Total Credit Hours: 134 Total Courses: 47 Semesters: 8 Duration: 4 Years	Category: Courses (Credit Hours) General Education: 15 (34) Major courses: 18(58) +7 (21) Allied/interdisciplinary courses: 5 (15) Internship/field experience: N/A Capstone project: 2 (6) Total: 47 (134)
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SCHEME OF STUDIES

S #	COURSE CODES	CLASS	COURSE TITLE	CREDIT HOURS	PRE-REQ	CO-REQ
SEMESTER 1						
1	CMC111	CORE	PROGRAMMING FUNDAMENTALS	3 + 0	-	CMC111-L
	CMC111-L	CORE	PROGRAMMING FUNDAMENTALS (Lab)	0 + 1		
2	GER111	GER	APPLICATION OF INFORMATION & COMMUNICATION TECHNOLOGIES	2 + 0	-	GER111-L
	GER111-L	GER	APPLICATION OF INFORMATION & COMMUNICATION TECHNOLOGIES (Lab)	0 + 1		
3	GER152	GER	NATURAL SCIENCE (APPLIED PHYSICS)	2 + 0	-	GER152-L
	GER152-L	GER	NATURAL SCIENCE (APPLIED PHYSICS (Lab))	0 + 1		
4	GER131	GER	CALCULUS AND ANALYTIC GEOMETRY	3 + 0	-	
5	GER121	GER	FUNCTIONAL ENGLISH	3 + 0	-	
				13+3(16)		
SEMESTER 2						
1	CMC112	CORE	OBJECT ORIENTED PROGRAMMING	3 + 0	CMC111	CMC112-L
	CMC112-L	CORE	OBJECT ORIENTED PROGRAMMING (Lab)	0 + 1	CMC111	
2	GER132	GER	DISCRETE STRUCTURES	3 + 0	GER131	
3	CMC123	CORE	DIGITAL LOGIC DESIGN	2 + 0	GER152	CMC123-L
	CMC123-L	CORE	DIGITAL LOGIC DESIGN (Lab)	0 + 1	GER152	
4	GER122	GER	EXPOSITORY WRITING	3 + 0	GER121	
5	MTE111	MATHS	MULTIVARIABLE CALCULUS	3 + 0	GER131	
6	GER241	GER	Pakistan Studies	2 + 0		
7	GEN111*	GER	Understanding of the Holy Quran-	0 + 1		
				16+3(19)		

SEMESTER 3						
1	CMC251	CORE	DATA STRUCTURES	3 + 0	CMC112	CMC251-L
	CMC251-L	CORE	DATA STRUCTURES (Lab)	0 + 1	CMC112	
2	CMC223	CORE	COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE	2 + 0	CMC123	CMC223-L
	CMC223-L	CORE	COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE (Lab)	0 + 1	CMC123	
3	AIC211	DOMAIN CORE	ARTIFICIAL INTELLIGENCE	2+ 0	CMC112	AIC211-L
	AIC211-L	DOMAIN CORE	ARTIFICIAL INTELLIGENCE (Lab)	0+ 1	CMC112	
4	CMC262	CORE	COMPUTER NETWORKS	2 + 0	CMC123	CMC262-L
	CMC262-L	CORE	COMPUTER NETWORKS (Lab)	0 + 1	CMC123	
5	MTE212	MATHS	PROBABILITY & STATISTICS	3 + 0	MTE111	
6	GEN 112*	GER	Understanding of the Holy Quran-II	0 + 1	GEN111	
				12+5(17)		
SEMESTER 4						
1	CMC242	CORE	OPERATING SYSTEMS	2 + 0	CMC251	CMC242-L
	CMC242-L	CORE	OPERATING SYSTEMS(Lab)	0 + 1	CMC251	
2	AIC212	DOMAIN CORE	PROGRAMMING FOR AI	2 + 0	AIC211	AIC212-L
	AIC212-L	DOMAIN CORE	PROGRAMMING FOR AI (Lab)	0 + 1	AIC211	
3	MTE213	MATH	LINEAR ALGEBRA	3 + 0	MTE212	
4	CMC252	CORE	ANALYSIS OF ALGORITHMS	3 + 0	CMC251	
5	MTE221	MTE	TECHNICAL & BUSINESS WRITING	3 + 0	GER122	
6	GER142	GER	IDEOLOGY AND CONSTITUTION OF PAKISTAN	2 + 0	-	
				15+2(17)		
SEMESTER 5						
1	CMC371	CORE	SOFTWARE ENGINEERING	3 + 0	CMC252	
2	CMC363	CORE	INFORMATION SECURITY	2 + 0	CMC262	CMC363-L
	CMC363-L	CORE	INFORMATION SECURITY (Lab)	0 + 1	CMC262	
3	CMC331	CORE	DATABASE SYSTEMS	3 + 0	CMC241	CMC331-L
	CMC331-L	CORE	DATABASE SYSTEMS (Lab)	0 + 1	CMC241	
4	XXXXXX **	DOMAIN ELEC	Domain Elective 1	3**	-	
5	AIC221	DOMAIN CORE	INTRODUCTION TO MACHINE LEARNING	2 + 0	AIC211	AIC221-L
	AIC221-L	DOMAIN CORE	INTRODUCTION TO MACHINE LEARNING (Lab)	0 + 1	AIC211	
6	GERXXX	GER	SOCIAL SCIENCES (ELECTIVE I) (GER261 - Intro to Mgmt.)	2 + 0	GER122	
				15+3(18)		

SEMESTER 6						
1	AIC323	DOMAIN CORE	ARTIFICIAL NEURAL NETWORKS & DEEP LEARNING	2 + 0	AIC221	AIC323-L
	AIC323-L	DOMAIN CORE	ARTIFICIAL NEURAL NETWORKS & DEEP LEARNING (Lab)	0 + 1	AIC221	
2	AIE423	DOMAIN CORE	COMPUTER VISION	2 + 0	AIC221/ AIC211	AIE423-L
	AIE423-L	DOMAIN CORE	COMPUTER VISION (Lab)	0 + 1	AIC221/ AIC211	
3	XXXXXX **	DOMAIN ELEC	Domain Elective 2	3**	-	
4	ESCXXX	ESC	SOCIAL SCIENCE II (ESC311 - Intro. To Mark.)	3 + 0	-	
5	AIC331	DOMAIN CORE	KNOWLEDGE REPRESENTATION & REASONING	2 + 0	CMC331	AIC331-L
	AIC331-L	DOMAIN CORE	KNOWLEDGE REPRESENTATION & REASONING (Lab)	0 + 1	CMC331-L	
6	GER141	GER	ISLAMIC STUDIES	2 + 0		
				13+4(17)		
SEMESTER 7						
1	CMC491	CORE	FINAL YEAR PROJECT - I	0 + 3	CMC371 / AIC323	
2	XXXXXX **	DOMAIN ELEC	Domain Elective 3	3**	-	
3	XXXXXX **	DOMAIN ELEC	Domain Elective 4	3**	-	
4	XXXXXX **	DOMAIN ELEC	Domain Elective 5	3**	-	
5	CMC443	CORE	PARALLEL & DISTRIBUTED COMPUTING	2+ 0	CMC241/ AIC323	CMC443-L
	CMC443-L	CORE	PARALLEL & DISTRIBUTED COMPUTING (Lab)	0 + 1	CMC241/ AIC323	
6	GER464	GER	ENTREPRENEURSHIP	2 + 0	-	
				10+7(17)		
SEMESTER 8						
1	CMC492	CORE	FINAL YEAR PROJECT - II	0 + 3	CMC491	
2	GER463	GER	PROFESSIONAL PRACTICES	2 + 0	-	
3	GER443	GER	CIVICS AND COMMUNITY ENGAGEMENT	2 + 0	-	
4	XXXXXX **	DOMAIN ELEC	Domain Elective 6	3**	-	
5	XXXXXX **	DOMAIN ELEC	Domain Elective 7	3**	-	
				8+5(13)		
			TOTAL CREDIT HOURS	134		

*Campus will decide to offer a 2 credit hrs. Course for non-Muslim students.

** The offering of the 3+0 or 2+1 electives will depend on the availability of the teaching and lab resources

 Change in credit hrs.

 New course added

 Move to align with other programs and Fest policy

 Name Change



List of Domain Elective Courses for BS AI Program

List of Domain Elective Course(s) (3+0)			
S.No.	Course Code	Course Title	Credit Hours
1.	AIE311	Advance Statistics	3+0
2.	AIE312	Expert Systems	3+0
3.	AIE313	Fuzzy Systems	3+0
4.	AIE314	Knowledge Based Systems	3+0
5.	AIE321	Data Warehouse and Data Mining	3+0
6.	AIE331	AI Agents	3+0
7.	AIE341	Deep learning	3+0
8.	SEE322	Agile Software Management	3+0

List of Domain Elective Course(s) (2+1)			
S.No.	Course Code	Course Title	Credit Hours
1.	AIE421	Data-driven Decision Making	2+1
2.	AIE431	Swarm Intelligence	2+1
3.	AIE441	Natural Language processing NLP	2+1
4.	AIE442	Generative AI	2+1
5.	CSE487	MLOps	2+1
6.	AIE444	Speech and Language Processing	2+1
7.	AIE445	Large Language Models (LLM)	2+1
8.	AIE461	Autonomous Robotics	2+1
9.	AIE471	Edge AI	2+1
10.	CSE384	Data Science	2+1
11.	CYE411	Vulnerability Assessment & Reverse Engineering	2+1
12.	CSC382	HCI & Computer Graphics	2+1
13.	MGE454	Virtual and Augmented Reality	2+1
14.	MGC223	Game Design and Development	2+1

Social Science Electives				
S. No.	Course Code		Title	Cr. Hr
1	GER262	ESC	Sociology	2 + 0
2	GER261	ESC	Introduction to Management	2 + 0
4	ESC311	ESC	Introduction to Marketing	3 + 0
5	ESC212	ESC	Phycology	2 + 0
6	ESC312	ESC	Behavioral Sciences	3 + 0
7	GER263	ESC	Economics	3 + 0
8	GER361	ESC	Project Management	2 + 0